

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	(a)		Substitution: $22\,000 = 30 \times \text{time}$ (1) $\text{Time} = \frac{22000}{30}$ (1) $= 733.33$ [h per year] (1) $= \frac{733.33 \text{ ecf}}{365} = 2$ [h] (1) accept 2.01 or 2.009 [h] Alternative: $\frac{22000}{365} = 60.27$ (1) Substitution: $60.27 \text{ ecf} = 30 \times \text{time}$ (1) $\text{Time} = \frac{60.27}{30}$ (1) $\text{Time} = 2$ [h] (1)	1	1 1 1		4	4	
	(b)		Energy saved = $22\,000 - 20\,500 = 1500$ [kWh] (1) Cost savings = $1500 \times 12 = 18\,000$ [p] (1) ecf on attempted energy saved subtraction shown Conversion to £180 ecf (1) Payback time = $\frac{3600}{180 \text{ ecf}} = 20$ [years] (1) Award 3 marks for an answer of 0.2 [years] Award 2 marks for an answer of 1.46 [years] or 1.36 [years]		4		4	4	
			Question 5 total	1	7	0	8	8	0